

Homework 02

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Problem 1. Let X , Y and Z be random variables and let c be a number. Prove that

$$\text{Var}(cX) = c^2 \text{Var}(X),$$

$$\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$$

and

$$\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z).$$

Problem 2. Let X_1 and X_2 be discrete random variables and suppose that a_1 and a_2 are real numbers. Using the joint probability mass function for X_1 and X_2 , show that

$$\mathbb{E}[a_1X_1 + a_2X_2] = a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2].$$

Problem 9. (Edition 11, Chapter 2.) If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0, \\ \frac{1}{2} & 0 \leq b < 1, \\ \frac{3}{5} & 1 \leq b < 2, \\ \frac{4}{5} & 2 \leq b < 3, \\ \frac{9}{10} & 3 \leq b < 3.5, \\ 1 & b \geq 3.5, \end{cases}$$

calculate the probability mass function of X .

Problem 12. (Edition 11, Chapter 2.) On a multiple-choice exam with three possible answers for each of the five questions, what is the probability that a student would get four or more correct answers just by guessing?

Problem 22. (Edition 11, Chapter 2.) If a fair coin is successively flipped, find the probability that a head first appears on the fifth trial.

Problem 37. (Edition 11, Chapter 2.) Let $X_1, X_2, \dots, X_n$ be independent random variables, each having a uniform distribution over $(0, 1)$. Let $M = \text{maximum}(X_1, X_2, \dots, X_n)$. Show that the distribution function of M , $F_M(\cdot)$, is given by

$$F_M(x) = x^n, \quad 0 \leq x \leq 1.$$

What is the probability density function of M ?

Problem 61. (Edition 11, Chapter 2.) Let X and W be the working and subsequent repair times of a certain machine. Let $Y = X + W$ and suppose that the joint probability density of X and Y is

$$f_{X,Y}(x,y) = \lambda^2 e^{-\lambda y}, \quad 0 < x < y < \infty.$$

1. Find the density of X .
2. Find the density of Y .
3. Find the joint density of X and W .
4. Find the density of W .

Problem 86. (From Edition 11.) Each new book donated to a library must be processed. Suppose that the time it takes a librarian to process a book has mean 10 minutes and standard deviation 3 minutes. If a librarian has 40 books that must be processed one at a time,

1. approximate the probability that it will take more than 420 minutes to process all these books;
2. approximate the probability that at least 25 books will be processed in the first 240 minutes.